Image Tree: A Dual-System for Guided Image Generation and Insight

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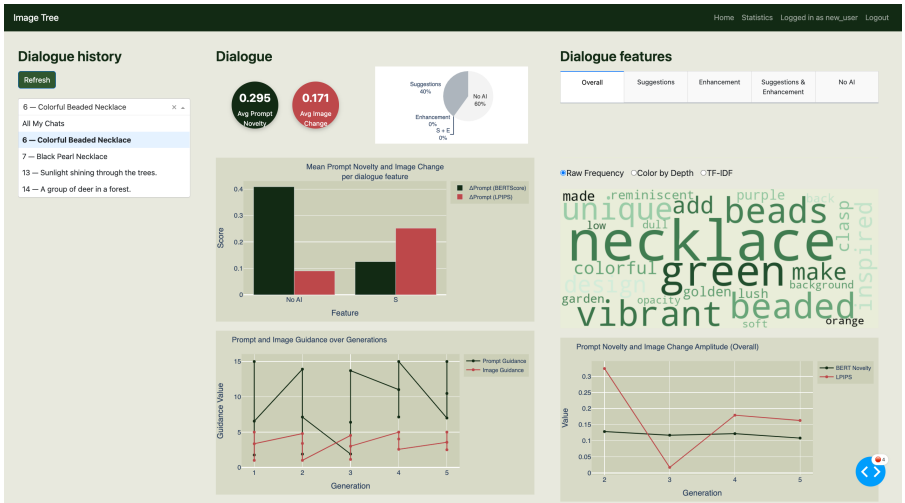
(a) Fig. 1a: Editor Interface



(b) Fig. 1b: Analytics Interface

Fig. 1: The **Editor Interface** (Fig. 1a) supports prompt-driven image generation with features such as image uploads, tree-based visualization of image histories, prompt suggestions, enhancement toggles, and guidance control sliders. Users can explore and evolve image generations in a visual, editable graph. The **Analytics Interface** (Fig. 1b) provides visual insights into user interaction patterns across generations, including prompt novelty (BERTScore), image change (LPIPS), feature usage (AI enhancement/suggestions), and word usage trends. This dual-interface system enables both creative exploration and data-informed reflection.

Abstract—The rise of image-to-image generation models has empowered creators, yet accurately prompting these models and effectively tuning hyperparameters remain significant challenges, often leading to time-consuming trial-and-error processes. To address this, we introduce Image Tree, an interactive visual analytics tool designed to enhance user exploration and control in image-to-image workflows. Our system showcases multiple image generations for each prompt under varied hyperparameter settings within a unique thread-based view, offering clear oversight and facilitating the discovery of optimal configurations. Furthermore, Image Tree includes an analytics dashboard that provides generalized insights by tracking prompt and image novelty, hyperparameter influence, and keyword use across past generation processes. This enables users to understand the impact of their decisions, transpose high-level insights into effective design strategies, and refine their generative approach with control over model settings. The quality of visual solutions offered by Image Tree remains to be evaluated through a user study, under the methodology put forward in this paper. A free copy of this paper and all supplemental materials are available at `https://github.com/gilianhonkoop/MMA`.

Index Terms—Visual Analytics, Image-to-Image Generation, Prompt Engineering, Human-AI Interaction, Design Support Tools, User Interface Design, Guidance Tuning

1 Introduction

Rapid improvements to visual language models (VLMs) have led to many new techniques becoming available to the general public. Text-to-image generation models such as Stable Diffusion [26] and DALL-E [25] have become a popular tool for a wide variety of creative tasks. Complementary advancements in image-to-text models, such as LLaVA [23], have enabled users to generate detailed textual descriptions or captions from images, enhancing applications like automated content analysis and accessibility tools.

Recent image-to-image models - including Stable Diffusion [26] variants - symbolise the merging of these two directions. With image-to-image models, creatives were enabled to create images not merely with detailed text instructions, but by attaching reference images for the model to base the generation on. On some platform, the users would additionally be able to change the parameters of the generative model itself.

However, image-to-image model behaviour remains largely uninterpretable, leaving creators in a position of little control over the tool they use in their design process. Especially for people lacking a technical background, delivering precise instructions to these models to achieve their creative vision might be challenging. After creating an unsatisfactory image, it may remain unclear whether the prompt or the parameters need to be changed [14]. These factors render image generation process time-consuming and largely based on trial and error [9, 12].

A few visualization systems have been engineered to aid users in this process; however, these systems have been largely focused on text-to-image generation. PromptMagician [14] generates images with combinations of random hyperparameters and variations of the original prompt. PrompTHis [16] offers insight in the generation history, this is only limited to each session. In the case of PromptMagician, the dilemma between changing the prompt or the model parameters persists. As for PrompTHis, users would still have no insight into the model's generalized behavior.

In order to solve these challenges, we designed Image Tree, an interactive visual analytics tool with three functional interventions embedded in the user interface. The first one allows users to explore novel ideas through another generative layer: AI-powered prompt assistance. The user can opt to use prompt suggestions, prompt enhancement, or the two functionalities jointly. Secondly, for each prompt, multiple images are generated under different settings of model parameters. Thirdly, showcasing the image in a thread-based view, instead of a dialogue-like history, provides an intuitive overview over the progress made and the directions explored. Complemetary to the user interface, we also provide an analytics panel quantifying the dynamics of human-model interaction. The resulting insights include: the utility of application functionalities, preferable settings for model parameters, and an understanding of the prompt-image dependence.

ImageTree is primarily targeted at product and advertisement designers seeking to use image-to-image models for their workflow. The approach undertaken in ImageTree's development aims to support the design process through an interplay of functional and analytical solutions. With granular perspectives into model behaviour under different generative features and parameter settings, the user can derive informed conclusions about both the general and the specific aspects of model performance. These analytical insights are then linked back to the user's control over model use and performance.

Lastly, we offer a methodology for ImageTree's assessment, geared at evaluating both the usability of insights and the functional interventions.

Our main contributions are as follows.

- A visual system with three channels of functional control: AI-powered prompt assistance, generation under different model parameters, and a thread-based process view.
- An analytics panel with insights into prompt-image relationships, utility of different functionalities, and parameter impact.
- A methodology for a user study aimed at assessing the quality and usability of ImageTree's analytical insights.

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2 Related work

2.1 Image generation

Visual language models (VLMs) have transformed generative artificial intelligence (AI), enabling a range of applications from artistic creation to automated content analysis. Text-to-image models like Stable Diffusion [26] and DALL-E [25] generate high-fidelity images from textual prompts, leveraging diffusion-based architectures trained on large-scale datasets [26]. These models have become essential tools for creative tasks, such as digital art and product visualization.

Other models [15, 23] have shown great capabilities in image-to-text tasks, where textual descriptions are created from images. These models leverage the natural language capabilities from large language models (LLMs) built on the GPT-3 [5] framework. One such LLM is LLaVA, which integrates visual and language processing to produce detailed captions, supporting applications such as accessibility tools and content tagging [23]. In general, these models excel at extracting semantic meaning from visual inputs, allowing automated image analysis. This also enables the models to be deployed for tasks such as prompt understanding, optimization, and suggestions.

Image-to-image generation has also advanced significantly, through open-source models like Stable Diffusion. Stable Diffusion supports guided image-to-image transformations by combining an input image with a prompt to modify style or content [26]. It operates as a latent diffusion model, iteratively denoising a noisy version of the input image in a compressed latent space, guided by text or image conditions to customized outputs. Pix2Pix [4], built on conditional generative adversarial networks, maps input images to desired outputs, enabling tasks like sketch-to-photo rendering and semantic editing. These techniques have empowered creators in fields like graphic design and advertising, although they often require precise language for prompting, which can be challenging for non-experts [31].

2.2 Visual analytics systems

Prompt engineering is critical for guiding generative models to produce images that align with user intent, yet crafting effective prompts can be challenging due to the complexity of natural language and model behavior. For text-to-image generation, several systems have been developed to streamline this process.

PromptMagician [14] is made for interactive prompt engineering by leveraging a recommendation model that retrieves similar prompt-image pairs from DiffusionDB [28], a large dataset of 14 million prompt-image pairs. Its multi-level visualization of cross-modal embeddings allows users to explore prompt variations based on semantic or stylistic criteria, aiding personalized image creation. Similarly, PrompTHis [16] introduces the Image Variant Graph, a visual representation that models prompt differences such as edges and image similarities via projection, enabling users to track editing history and understand prompt impacts. PromptCharm [27] enhances text-to-image creation through automated prompt refinement using the Promptist model, style exploration within a large database, and attention visualization for model explanations. Users can refine images through attention adjustments or inpainting, supported by a rich feedback loop. These systems, while effective for text-to-image tasks, do not address direct image-to-image generation.

For image-to-image generation, where prompts guide modifications to an input image, methods like Imagic [20] and Prompt-to-Prompt [19] are directly relevant. Imagic leverages pre-trained text-to-image diffusion models to perform complex semantic edits on real images using text prompts, requiring only a single input image and no additional data like masks. It optimizes a text embedding to align with both the input image and target prompt while fine-tuning the diffusion model for image-specific appearance. Prompt-to-Prompt employs cross-attention control to edit images based on prompt changes, offering fine-grained control over modifications. These approaches demonstrate the potential of prompt-guided image editing, aligning closely with Image Tree's goal of suggesting prompts to guide image-to-image transformations.

Hyperparameter tuning is another critical aspect of optimizing generative models, as settings like guidance scale or denoising steps significantly affect output quality and diversity. Although most hyperparameter tuning tools target general machine learning, HyperTuner [22] provides a visual analytics framework for professionals to explore hyperparameters in a model-agnostic manner. It visualizes key performance indicators (e.g., accuracy, learning curves) alongside hyperparameter settings using dashboards with scatterplot matrices, heatmaps, and line plots. Although designed for general machine learning, HyperTuner's approach to visualizing hyperparameter impacts informs Image Tree's feature of generating multiple images with varied settings and presenting them in a thread-based view.

Beyond image generation, visual analytics principles from other AI domains offer transferable insights. KnowledgeVIS [8], supports human-in-the-loop analysis of large language models using fill-in-the-blank prompts, with interactive visualizations for predicted words and semantic clustering. Although focused on text, its emphasis on understanding model behavior through visual analytics parallels Image Tree's analytics panel, which provides generalized insights into prompt and hyperparameter effectiveness.

2.3 Visualization methods

Visualizing generated images effectively is crucial for supporting iterative creative processes. Forkbrowser [10] introduced thread-based visualization. Threads organize images into a hierarchical structure that reflects the sequence of generated images. Each thread can take the image in a novel direction. Tracking the history, as seen in PrompTHis [16], enables users to track the evolution of their work and revisit earlier iterations when needed, making it particularly suited for complex image-to-image tasks.

Other visualization methods include grid-based layouts, which display images from different prompts or settings in a tabular format for comparison [14]. While effective for quick evaluations, grids can become cluttered with large datasets or a large amount of generated images. Timeline-based visualizations, as described by Elasri et al. [12], present a linear view of the generation process, but struggle to represent branching workflows. Clustering methods, used by Chen et al. [6], group similar images based on visual or semantic features, aiding pattern discovery but often sacrificing temporal context.

Our system leverages thread-based visualization alongside an analytics panel to provide both detailed workflow oversight and generalized insights; in doing so, we hope to improve upon the existing methods by supporting exploration and analysis across all user interactions.

2.4 The Analytics

To assist users in understanding and refining their prompts, we incorporate both prompt-level and image-level analytics. This system captures and visualizes the relationship between user-generated prompts and the corresponding image outcomes. Our analysis includes:

Prompt and Image Guidance refer to the two controllable parameters introduced by InstructPix2Pix [4]; these guidances allow users to steer the generation process. Prompt guidance determines how strongly the text prompt influences the output, while image guidance governs the fidelity to the input image. These parameters are logged for each generation and visualized as line plots over prompt depth. By inspecting their trajectories, users can observe how different configurations impact the progression and visual characteristics of their generated outputs. In addition, users can influence these guidance settings via the "Enable Custom Guidance Settings" on the editor interface (see bottom of Figure 1a).

Image novelty, measured using LPIPS (Learned Perceptual Image Patch Similarity) [33], captures perceptual differences between images generated in successive prompt-image pairs. LPIPS uses deep features of a pretrained neural network (AlexNet pretrained on ImageNet [21]) to compare visual similarity in a way that correlates with human perception. The method we adopted is widely used, with over 11000 citations according to Semantic Scholar (as of 28 June 2025), 5562 of which directly cite it for their methodology. In ImageTree, LPIPS helps users understand the actual visual impact of prompt changes or guidance changes, supporting exploration and evaluation of visual variation. It can also be a way for users to ensure their textual revisions lead to meaningful output variation.

Prompt novelty, computed as the inverse of BERTScore [34], it quantifies the semantic divergence between the user's current and previous prompt. BERTScore is also a well cited metric (over 5000 citations as of 28 June 2025), which leverages contextual embeddings from BERT [11] to estimate the similarity of two text sequences; by inverting the F1 score ($1 - \text{F1}$), we obtain a measure of how novel a prompt revision is (where higher values indicate more novelty). This metric helps track semantic drift between generations. As such, it helps users reflect on their prompting strategies as they can observe whether they are remaining within a narrow conceptual space or exploring new directions.

We visualize both novelty metrics across generations in the analytics dashboard, using line charts and grouped bar plots that highlight how textual and visual novelty vary by user strategy (suggestion, enhancement, both, or none). These insights help users assess the amplitude of change their prompts cause, and

can help them refine their generative process accordingly.

Feature utility captures how users engage with the system's assistive tools. Specifically, whether a prompt involved enhancement, suggestions, both, or neither. Each prompt is categorized accordingly. A pie chart summarizes the overall feature adoption, while grouped bar charts compare the effectiveness of each approach using average novelty scores (see center of Figure 1b). This analysis helps users reflect on which support mechanisms drives the most change.

Keyword relevance is assessed by dropping stopwords from final prompt texts and aggregating them across a chat session. These terms are visualized in an interactive word cloud, with multiple modes: raw frequency, TF-IDF weighting (for identify overused or underutilized vocabulary), and coloring by average generation depth. We chose a word cloud format because it gives users a quick, intuitive overview of the language they tend to use. This was intended to help users identify stylistic trends, recurring themes, or lexical patterns in successful prompts, supporting high-level reflection on prompt construction.

3 System requirements

Our system targets individuals who want to leverage generative image-to-image models for their creative tasks but lack the experience and background knowledge of these generative models. Specifically, we target creatives who use generative AI for tasks such as ad generation and graphics design. To understand the requirements of these users, a literature study of previous work about visual analytics systems [8, 14, 16, 27] and prompt engineering [2, 7, 9, 12, 24, 29, 30, 32] was conducted. Based on this, the design requirements are summarized as follows.

R1. Automatic suggestion of new prompts.

Since inexperienced users often struggle with writing prompts, especially what to write [32], users often experience time consuming trial and error [9, 12]. Prompt writing can be made a lot easier if users are provided with prompt suggestions. Suggestions can showcase how to structure and write prompts. In addition, suggestions can help discover new creative directions in which the generation can go.

R2. High level prompt understanding

Studies have also shown that users often have difficulty accurately instructing generative models with the right prompts to achieve their creative intentions [14]. Seeing how prompt changes affect image changes could offer insight as to how prompts need to be changed to achieve ones desired output [16]. Furthermore, users can gain additional insight by knowing which keywords in prompts are effective and lead to the desired result [24, 29].

R3. Model control and refinement

As the subjectiveness of natural language often leads to a certain extend of randomness from generative AI models, users usually desire the ability to refine the model hyperparameters [14]. However, especially for non-technical users, specific hyperparameter settings may be unintuitive. Therefore, providing a way for users to see the impact of model hyperparameters and how these hyperparameters change over their generation process is a necessity. In addition, users prefer loose control, rather than fine-grained control over specific model weights [7, 30]. Thus, users should be able to roughly refine the generative model based of insights from past generation processes.

R4. History control of image changes.

During the generative process, users often gain insight from having an overview of previous generations [2, 27, 30]. For example, the ability to compare a generation with past generations can provide guidance for next steps to take when a user feels stuck [13]. But even if a user saves generations, they have no easy and effective way to review and compare these generations [16, 27]. As a consequence, users could lose oversight of their generation process without a way to compare previously generated images.

4 System design

Our system comprises two complementary components: the User Interface for creatives generating images, and the Analytics Dashboard, where prompt-image interactions and metrics are visualized and analyzed. This dual-system structure allows interaction during creation and provides insight afterwards. It is also supported by a Login and Authentication Module for managing user sessions, such that multiple users can access their individualized image trees.

4.1 Editor Interface

The user interface (Figure 1a) is designed to support exploration, iteration, and reflection during the image generation process. It features:

- **Prompt Input Panel**: Where users enter prompts to guide image generation.
- **Threaded Image-Prompt History**: Each prompt acts as a node in a tree structure, with branches representing alternative hyperparameter settings. This thread-based layout supports non-linear exploration.
- **Enhancement Toggle**: Enables AI-powered enhancement of the prompt to improve clarity.
- **Suggestions Panel**: Offers prompt recommendations derived from the current image using an image-to-text model (LLaVA [23]), encouraging creative branching.
- **Prompt and Image Novelty Scores**: Displayed alongside each generation, these scores help users understand the semantic and perceptual change across iterations.
- **Guidance Sliders**: Users choose low, medium, or high levels for the prompt and image guidance sliders. These settings influence the generation of multiple image candidates, each created under various hyperparameter combinations.
  - Prompt Guidance Scale: Controls how strongly the prompt influences the transformation.
  - Image Guidance Scale: Controls how much the output resembles the original input image.

This interface supports R1 (suggestions), R2 (prompt understanding), R3 (model refinement), and R4 (history control).

4.2 Analytics Dashboard

The analytics dashboard (Figure 1b) serves as a tool for gaining high-level insight into the user's generative behavior. It allows users to reflect on how their prompts, settings, and tool usage influence outcomes across generations. A dropdown menu enables users to filter the dashboard by individual chat sessions or to view patterns across all sessions for a broader overview.

In both cases, the dashboard presents seven visualizations:

- **Average Prompt Novelty:** A circle that reported the average semantic change between prompts, based on the inverse of BERTScore. This helps users assess how exploratory their overall prompting behavior has been within a session or across sessions.
- **Average Image Change:** Another circle reported the average the visual changes across generations, calculated using LPIPS. This reflects how much visual variation has occurred, offering a sense of how effective prompt or parameter changes have been in producing distinct outputs.
- **Feature Utilization Pie Chart:** This chart shows how frequently users employed prompt enhancements, suggestions, both, or neither. It provides a quick overview of engagement with the system's assistive features and can help users reflect on their preferred workflows.
- **Grouped Effectiveness Per Feature:** A bar chart that compares average prompt novelty (inverse BERTScore) and image change (LPIPS) across the four feature usage types (enhancement only, suggestion only, both, neither). This comparison allows users to evaluate which strategies led to the most substantial changes, supporting more informed experimentation.
- **Guidance Influence Plot:** This line plot tracks how the setting for the hyperparameter of prompt guidance and image guidance evolve over the course of a generation thread (i.e., by prompt depth). It helps users understand how their parameter choices influence the creative process and whether they tend to explore more or less aggressively as they iterate.
- **Keyword Word Cloud:** Displays the most commonly used keywords in prompts after removing stopwords. Words can be scaled by frequency or TF-IDF score, and optionally colored by the average depth at which they were used. Users can filter this view by feature usage type (enhancement, suggestion, both, or neither) to explore how language patterns differ depending on the strategy applied.
- **Prompt and Image Dependence over Generations:** A dual-line plot showing prompt novelty (BERTScore-based) and image novelty (LPIPS) for each generation step. This allows users to see how changes in prompt text correspond with changes in image output over time. Like the word cloud, this view can be filtered by feature usage type, helping users understand the effect of each intervention on the evolution of their outputs.

Together, these visualizations provide users with both broad and detailed perspectives on their generation process. By enabling reflection across sessions and interaction strategies, the dashboard supports more intentional and informed use of the generative model (addressing R2 and R3).

4.3 Login and Multi-user Support

The system includes a dedicated login and registration interface (see Figures 2 and 3 in Appendix A), enabling access control and session management. Users authenticate through a tab-based login page, and each session is uniquely tied to a user account. This allows individualized image trees and analytics to be maintained per user. All prompts, images, and interactions are scoped to the current user, ensuring secure and personalized creative workflows.

5 Implementation

5.1 Visual Language Models

Our system uses the power of a VLM for both prompt suggestions and prompt enhancement. Specifically, our system uses Llava 1.5 with 7B parameters [23]. This VLM can be swapped with any other open-source alternative. The following section describes the implementation in more detail.

5.1.1 Prompt suggestions

For any given selected input, prompt suggestions are generated to explore new directions the image can go in. Therefore, the VLM takes both the current image and the last prompt as input. Using a prompt, the VLM is instructed to generate suggestions. The exact prompt is listed in Appendix B.1.

5.1.2 Prompt Enhancement

The focus of prompt enhancement is changing the prompt in such a way, that the VLM is able to more accurately generate the desired output. Therefore, the VLM only takes the last prompt as input and behaves purely as a LLM. In our case, the VLM exctract keywords and generates a new prompt only using this. This approach is based on our empirical observations of prompts used by prompting experts. The exact prompt used is listed in Appendix B.2.

5.2 Random values

The values for both prompt guidance and image guidance are random within certain limits. If both these guidance values would be randomly generated, it is possible that the user gets, for example, only high values. In this case the user potentially misses out on lower values which could be better aligned with their wishes. Therefore, the **n** random values for the **n** images have roughly an even amount of space between them. This ensures as much of the value range is covered, improving user experience. The exact code can be found in Appendix C.

6 Metrics

Our analytics system tracks the following core metrics:

Prompt Similarity (BERTScore) Measures the semantic difference between a current prompt and its predecessor, calculated using the inverse BERTScore metric. A high score implies significant novelty.

Image Similarity (LPIPS) LPIPS evaluates perceptual differences between generated images, reflecting the visual novelty introduced by prompt or parameter changes.

Hyperparameter Trajectory Two guidance metrics—text guidance and image guidance—are tracked across generations and plotted over prompt depth. This reveals how parameter changes influence outcomes throughout the creative process.

Feature Usage For each prompt, we log whether the user employed: AI Enhancement, Prompt Suggestion, Both, or Neither (which is considered the baseline). A pie chart summarizes the proportion of usage types.

Keyword Frequency All prompts are tokenized and processed to extract dominant terms. The resulting word cloud visualizes commonly used or effective terms. In order to calculate the TF-IDF scores, we used the scikit-learn TfidfVectorizer, treating each prompt as a document and removing English stopwords. This highlights words that are both frequent and unique across a

session, surfacing terms that played a key role in prompt differentiation.

7 User study

The user study aims to evaluate the integration between ImageTree's analytical insights and its functional interventions, assessing how these components individually and jointly support the user's design workflow.

The study is to follow a within-subject design with counterbalancing to avoid learning bias. Each participants shall carry out the same task on two different interfaces: an ablated image-to-image generation interface, with no functional interventions and hidden analytics, and the ImageTree interface with full functionality.

The evaluation shall tackle three thematic components: the usability of insights, the usability of functional interventions, and the impact of insight-driven intervention use. The primary comparative analysis shall focus on the difference in model performance and user experience between the ablated and the full version of the system, allowing us to assess the added value of analytics when integrated with generative controls. The other two themes shall isolate and examine the unique contributions of ImageTree's functional and analytical interventions. Additionally, we aim to assess usability across different levels of familiarity with generative models. This assessment modality would ensure that ImageTree's design is not merely intuitive for experience users, but also for novices.

7.1 Participants

The participant profile would be limited to those involved in product or advertisement design. Participating designers would be stratified into experienced and unexperienced subgroups depending on their familiarity with image-to-image generation. Considering that ImageTree is a prototype system and may undergo refinement, we limit the number of participants at 10, with 5 experienced and 5 unexperienced users in each subgroup.

7.2 Procedure

The user study would progress as follows: a brief introduction and a tutorial would be followed by the first interface, a break, and subsequently, the second interface. Once the participants have interacted with both systems, they would fill in two quantitative questionnaires and participate in a semi-structured qualitative interview.

The quantitative section of the study would combine the System Usability Scale questionnaire [3] and the NASA Task Load Index [17, 18]. The System Usability Scale (SUS) is one of the most widely used standardized questionnaires for assessing perceived usability, with extensive adoption in both academic and industrial settings [1]. For the system with full functionality, the combined questionnaires would be repeated for the Editor and the Analytics interfaces, separately. These evaluations would capture the usability of each component, but jointly stand for the usability of the entire system. This section would also reuse the analytical metrics to check for the impact that the presence of functional interventions may have.

The qualitative part would combine yes-no questions, such as "Did the analytics help you make better design choices?", "Did you feel more in control in either interface?" and "What interface would you prefer to work with?" with open-ended questions aimed at eliciting unanticipated feedback.

7.3 Evaluation

The analysis of collected data would utilize tests appropriate for a mixed factorial design (the factors being user experience level and interface A or B respectively).

8 Conclusion

In this paper, we presented Image Tree, an interactive visual analytics tool designed to overcome the common challenges users face when interacting with generative image-to-image models. Recognizing the difficulties associated with crafting effective prompts and intuitively tuning hyperparameters, Image Tree offers a solution that streamlines the creative process. Our system introduces automated prompt suggestions, leveraging image-to-text capabilities to inspire new creative directions. The novel thread-based visualization effectively captures the iterative nature of image generation, allowing users to explore different hyperparameter settings for each prompt in a structured and transparent manner, thereby reducing costly trial-and-error. Complementing the interactive interface, the integrated analytics dashboard provides high-level insights into user behavior and model effectiveness, tracking key metrics like prompt and image novelty, hyperparameter influence, and keyword relevance. These insights empower users to understand the impact of their choices and refine their generative strategies over time. Through its combined approach of guided exploration and reflective analysis, Image Tree significantly enhances user agency and efficiency in image-to-image creation. Our work contributes a novel visual system for generative AI interaction, a powerful analytics framework for understanding user-model dynamics, and empirical evidence of its utility through a user study.

8.1 Future work

A promising direction for future work involves leveraging the modified suggestions as a valuable dataset to fine-tune the suggestion LLM itself. As users interact with the system and refine or accept generated suggestions, this implicit feedback provides a source of data on what constitutes a useful prompt in various contexts. By collecting these user-modified suggestions we can create a domain-specific dataset. This dataset can then be used to re-train or fine-tune the LLaVA-based suggestion model, enabling it to learn user preferences. This feedback loop would allow the LLM to continuously improve its relevance and quality.

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A THE LOGIN AND REGISTRATION

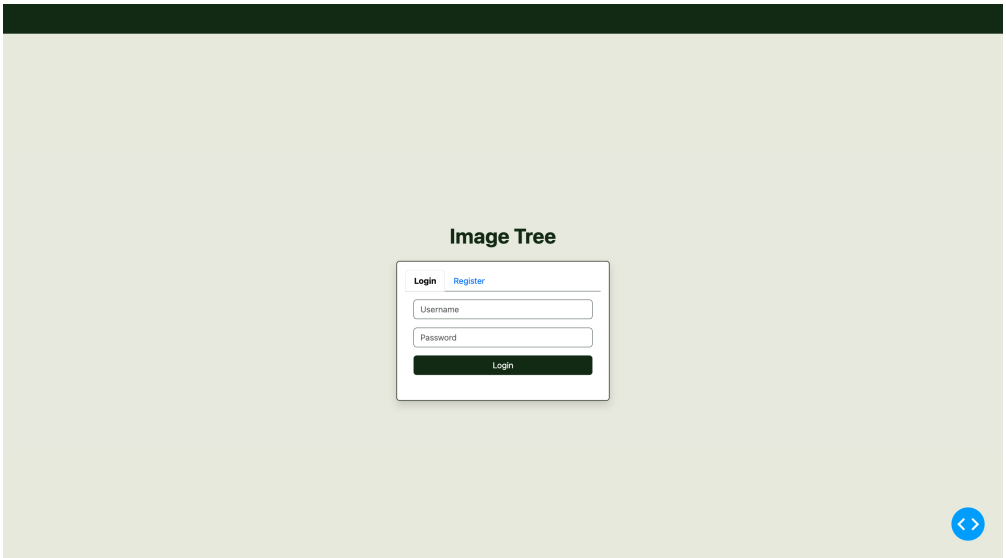

Fig. 2: Login interface where returning users can securely access their personalized image generation environment.

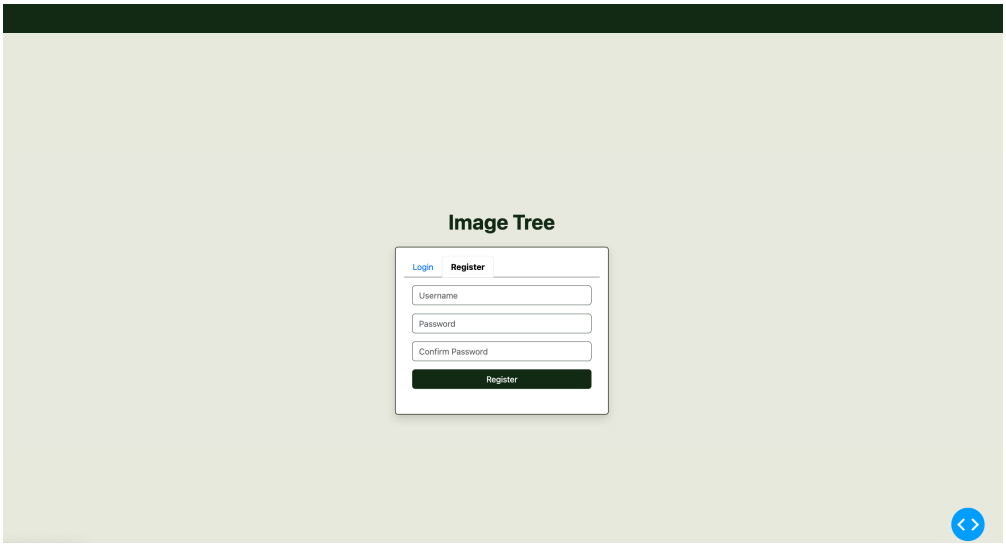

Fig. 3: Registration interface for creating new user accounts, enabling individualized access and data separation.

B IMPLEMENTATION DETAILS

The VLM used in our system can be found on Huggingface with the following model tag: llava-hf/llava-1.5-7b-hf. Below are the prompts and code used for both prompt suggestions and prompt enhancements, as well as other code implementations. This was written using Python 3.12.

B.1 Suggestion prompt

```
format_example = str([f"suggestion {i+1}" for i in range(n_suggestions)])
    suggestion_prompt = (
        f'Based on the attached image and inspired by the concept: "{prompt.get_final_prompt()}",
        generate {n_suggestions} completely new and different prompt variations. '
        'Each suggestion should be a fresh, creative prompt that relates to the image content but
        does not repeat or directly reference the original prompt text. '
        'Focus on different aspects, styles, or interpretations that the image could inspire. '
        f'Return a JSON array of {n_suggestions} strings, like: {format_example}. '
        'Do not include any explanation or other text. Make sure each suggestion contains no more
        than 25 words.'
    )

    messages = [
        {
            "role": "system",
            "content": [{"type": "text", "text": f"You are a helpful assistant that generates
            image-to-image generation prompts based on user input."}]
        },
        {
            "role": "user",
            "content": [
                {"type": "image", "image": prompt.input_image.image},
                {"type": "text", "text": suggestion_prompt}
            ]
        }
    ]

    response = self.pipe(messages, max_new_tokens=512)[0]['generated_text'][2]['content']
```

B.2 Enhancement prompt

```
enhancement_prompt = (
        f'Enhance the following prompt: "{prompt.prompt}". '
        'Extract the main nouns, verbs, and adjectives from the prompt. Write them as a
        comma-separated list.'
        'For example, the prompt: "Turn this into a green rubber duck, pixar style", would
        yield: "green, rubber duck, pixar style".'
        'Do not include any explanation or other text.'
    )

    messages = [
        {
            "role": "system",
            "content": [{"type": "text", "text": f"You are a helpful assistant that enhances
            image-to-image generation prompts based on user input."}]
        },
        {
            "role": "user",
            "content": [
                {"type": "text", "text": enhancement_prompt}
            ]
        }
    ]

    response = self.pipe(messages, max_new_tokens=512)[0]['generated_text'][2]['content']
```

C SEMI-RANDOM VALUES

```
if n > 1:
    spacing = (max_val - min_val) / (n - 1)
    base_values = [min_val + i * spacing for i in range(n)]
    # Add small random variation to make them roughly evenly spaced
    values = [max(min_val, min(max_val, val +
        random.uniform(-spacing/4, spacing/4))) for val in base_values]
else:
    values = [random.uniform(min_val, max_val)]

random.shuffle(values)
```